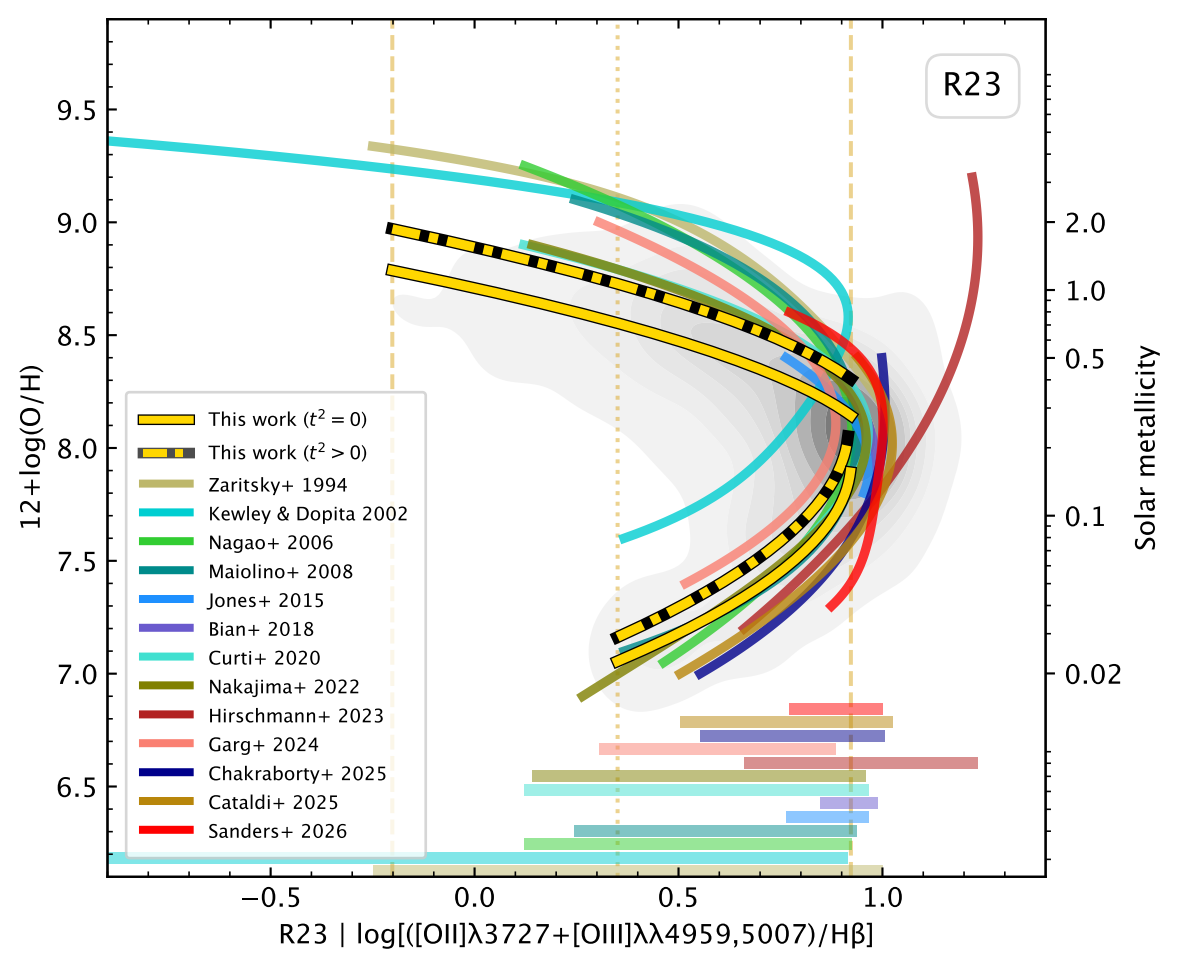

R23
This work ($t^2 = 0$)
This work ($t^2 > 0$)
Zaritsky+ 1994
Kewley & Dopita 2002
Nagao+ 2006
Maiolino+ 2008
Jones+ 2015
Bian+ 2018
Curti+ 2020
Nakajima+ 2022
Hirschmann+ 2023
Garg+ 2024
Chakraborty+ 2025
Cataldi+ 2025
Sanders+ 2026
12+log(O/H)
9.5
9.0
8.5
8.0
7.5
7.0
6.5
Solar metallicity
2.0
1.0
0.5
0.1
0.02
−0.5
0.0
0.5
1.0
R23 | log[([OII]λ3727+[OIII]λλ4959,5007)/Hβ]